Probabilistic (Graphical) Models

Part 3: Latent Variables, (Gaussian) Mixture Models, Expectation Maximization, Variational Inference

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'a'/'z' keys to browse the slides, 'o' for an overview, 'd' for dark/light mode

Objectives for the (P)(G)(M) Course

- Understanding
  - what is a probabilistic model
    - from fitting a distribution
    - to huge generative models
  - what is a Probabilistic Graphical Model (PGM)
  - what are the inference/learning methods
  - how this relates/differs/interacts with other approaches (e.g. deep learning, VAE, diffusion)
- Skills, be able to
  - model a problem with probabilities
  - reason using probabilities
  - read and write graphical models
  - select (and implement) models and inference methods

Syllabus for the PGM Course / Disclaimer

α. Introduction, the need for uncertainty

β. Probabilistic/generative modelling, maximum likelihood

γ. Probabilities, Bayesian principles and Bayesian learning

δ. Closed-form learning (conjugate priors)

ε. Inference Methods for Probabilistic (Latent Variable) Models

ζ. Monte Carlo Methods (Gibbs sampling, MCMC, HMC, NUTS)

η. Variational Auto-Encoders (and variational inference)

θ. Deep Learning meets Probabilistic Models

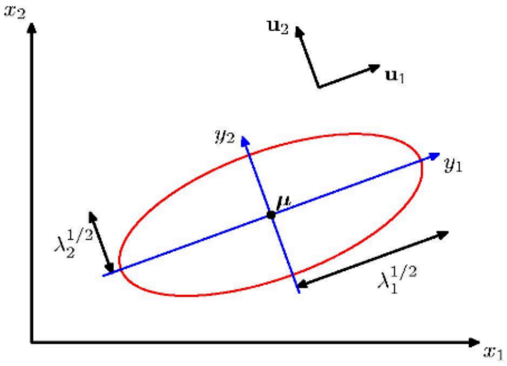
ι. Student Project

(Gaussian) Mixture Models (GMM)

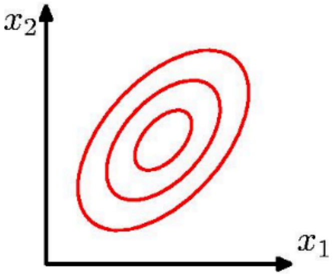
Aside: Multivariate Normal Distribution …

- D-dimensional space: $x = (x_1, ..., x_D)$

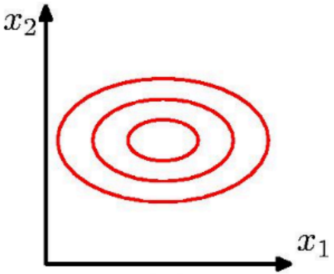
- Probability distribution
  - $\mathcal{N}(x|\mu,\Sigma) = \frac{1}{\sqrt{(2\pi)^D det(\Sigma)}} exp\left(-\frac{(x-\mu)^T\Sigma^{-1}(x-\mu)}{2}\right)$
  - Σ: covariance matrix (PSD[W], positive semidefinite)



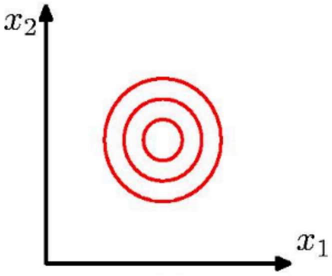
Elliptical equiprobability surfaces.
Major axes = eigenvectors of Σ



full cov. matrix
(½D(D+1) parameters)



diagonal cov. matrix
(D parameters)



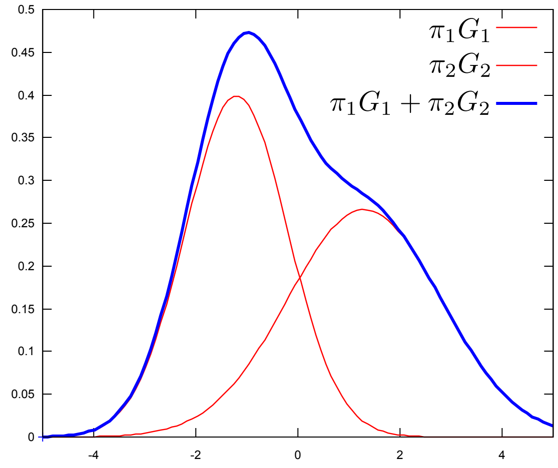
identity cov. matrix
(1 parameter)

Aside: Summing Densities vs Summing Random Variables

Considering two random variables and two derived random variables

- A normally distributed (10, 1)
  - $A \sim \mathcal{N}(10, 1)$
  - $f_A(a) \propto e^{-\frac{(a-10)^2}{2}}$ (density)
- B normally distributed (mean 40, standard deviation 2)
  - $B \sim \mathcal{N}(40, 2^2)$
  - $f_B(b) \propto e^{-\frac{(a-40)^2}{2\times 2^2}}$
- $C = A + B$
  - ✎
- $f_D = f_A + f_B$
  - ✎ ½

Gaussian Mixture Models (GMM)



- Weighted sum of Gaussians
- Parameters with K Gaussians
  - $\pi_{1..K}$: weights such that $\sum_{k=1}^{K} \pi_k = 1$
  - $\mu_{1..K}$: means of the Gaussians
  - $\sigma^2_{1..K}$: variances of the Gaussians
- Generative story (more on next slide)

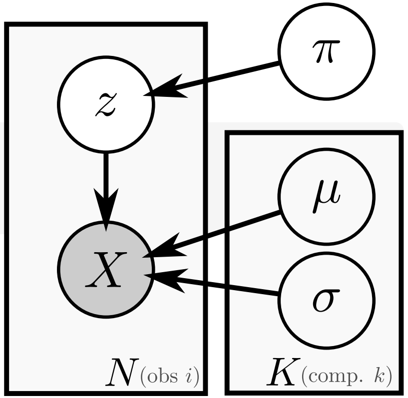
> for each point i, we first select a component index z_i according to the probability vector π, and then we sample the point X_i from this gaussian component $(\mu_{z_i}, \sigma_{z_i})$

(bayesian network? generative process?)

GMM Bayesian Network

- Generative story

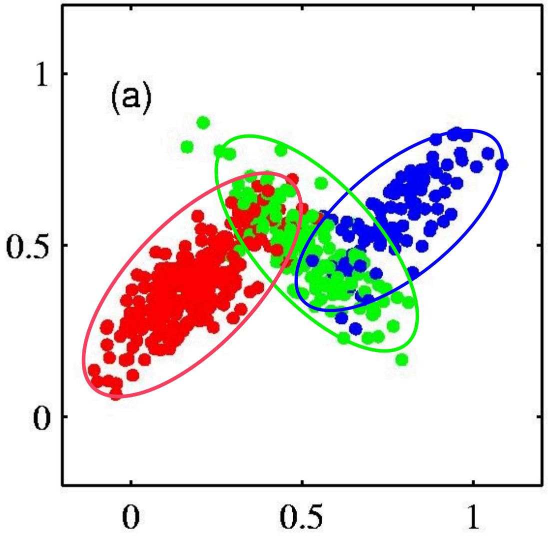
> for each point i, we first select a component index z_i according to the probability vector π, and then we sample the point X_i from this (gaussian) component $(\mu_{z_i}, \sigma_{z_i})$



- Generative process

$$
\begin{aligned}
& \pi \sim Prior_\pi() \\
\forall k \in [\![1, K]\!], \quad & \mu_k, \sigma_k \sim Prior_{(\mu,\sigma)}() \\
\forall i \in [\![1, N]\!], \quad & z_i \sim Categorical(\pi) \\
& X_i \sim Normal(\mu_{z_i}, \sigma^2_{z_i})
\end{aligned}
$$

- Example samples (in 2D)
  - $K = 3$ gaussian components
  - N draws
  - show the "*complete data*" (the color encodes the z_i)



Expectation Maximization for GMM

Learning with Expectation Maximization (EM)

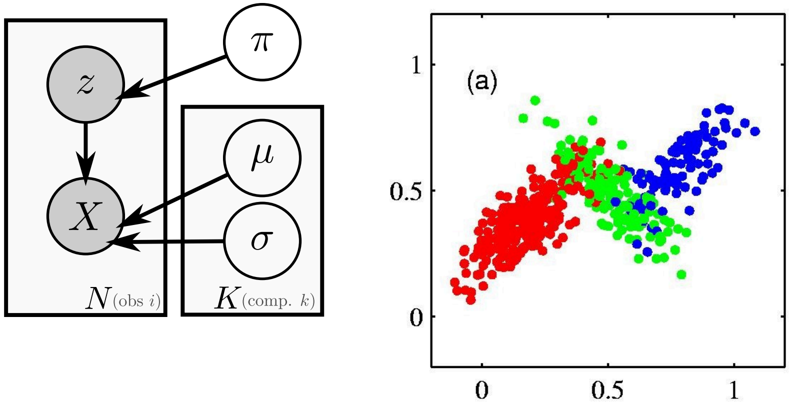
- Goal: given some observations, find the “best” parameters
  - parameters: $\theta = \{\pi_k, \mu_k, \sigma_k\}_{k=1..K}$
  - best = maximum likelihood estimator (MLE)
  - find $\theta_{ML} = \underset{\theta}{\operatorname{argmax}} \mathcal{L}(\theta|D) = \underset{\theta}{\operatorname{argmax}} p(X|\theta)$
  - $difficult$ due to latent variables, i.e. $p(X|\theta)$ vs $p(X, Z|\theta)$
- Log-likelihood function:

$$
\begin{aligned}
\ln \mathcal{L}(\theta|X) &= \ln p(X|\theta) = \ln \textstyle\prod_i p(X_i|\theta) \\
&= \textstyle\sum_i \ln p(X_i|\theta) = \sum_i \ln \displaystyle\sum_{k=1}^{K} p(X_i, z_i = k|\theta) \\
&= \textstyle\sum_i \ln \displaystyle\sum_{k} p(X_i|z_i = k, \theta) p(z_i = k|\theta)
\end{aligned}
$$

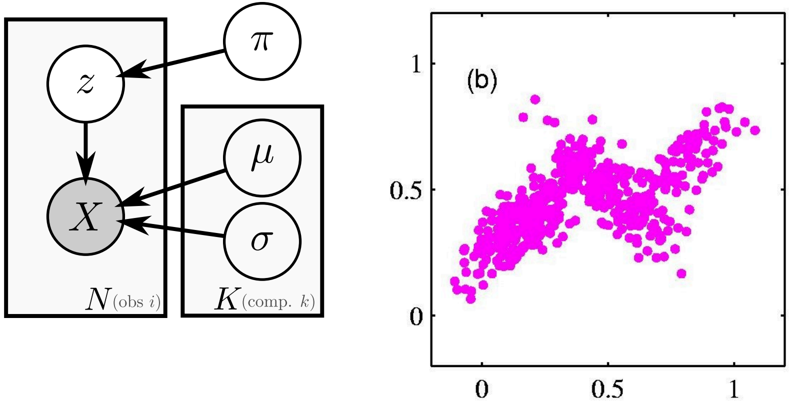
- Derive the closed-form MLE
- Problem
  - "Incomplete data", i.e., z unknown
  - i.e., $\sum_k$ in the likelihood ⇒ difficult to optimize

Complete/Incomplete Data: Illustration

- Complete data
  - supposing we know z
  - z: known component labels (each point: red, green or blue)
  - estimating θ is easy



- Incomplete data (actually observed)
  - “unsupervised” learning of the model
  - difficult to estimate θ
  - need to find z at the same time
  - use of Expectation Maximization algorithm



EM Intuition

- Complete data log-likelihood
  - $\ln \mathcal{L}_C(\theta|X, z) = \ln p(X, z|\theta)$
  - easier to maximize to *get* θ_{ML}
  - but need to know z
- If we knew θ_{ML} (but not z)
  - we could estimate the posterior of z, i.e., $p(z|X, \theta)$
  - i.e., how probable are the different values of z
  - i.e., for each point i and component k,
    $p(z_i = k|\ldots)$: the “responsibility” of component k for X_i

EM: The E and M steps

- Iterative algorithm
  - random initialization: $\theta^0 = rand()$
  - local optimum ⇒ needs multiple initializations
- E step
  - use the current estimate θ^{old}
  - to find the posterior $q_i(z_i) = p(z_i|X, \theta^{old}) \overset{indep.}{=} p(z_i|X_i, \theta^{old})$
    also named responsibility, i.e. $\forall k, R_{ik} = p(z_i = k|X_i, \theta^{old})$
- M step
  - use the computed $R_{ik} = p(z_i = k|X_i, \theta^{old})$ to find a new best estimate

$$
\begin{aligned}
\theta^{new} &= \operatorname{argmax}_\theta Q(\theta|\theta^{old}) \\
&= \operatorname{argmax}_\theta \text{“}\mathcal{L}(\theta|X, \theta^{old})\text{”} \\
&\overset{\text{def}}{=} \operatorname{argmax}_\theta \mathbb{E}_q \left[\ln p(X, Z|\theta)\right] \\
&= \operatorname{argmax}_\theta \sum_i \sum_k q_i(z_i = k) \ln p(X_i, z_i = k|\theta) \\
&= \operatorname{argmax}_\theta \sum_i \sum_k R_{ik} \ln p(X_i, z_i = k|\theta)
\end{aligned}
$$

What is it to be bayesian, reminder

Bayesian (people) do what?

1. use probabilities to represent (lack of) knowledge
2. have (explicit) priors on everything unknown
3. keep distributional estimate as long as possible/needed

Variational Inference, Variational Bayes

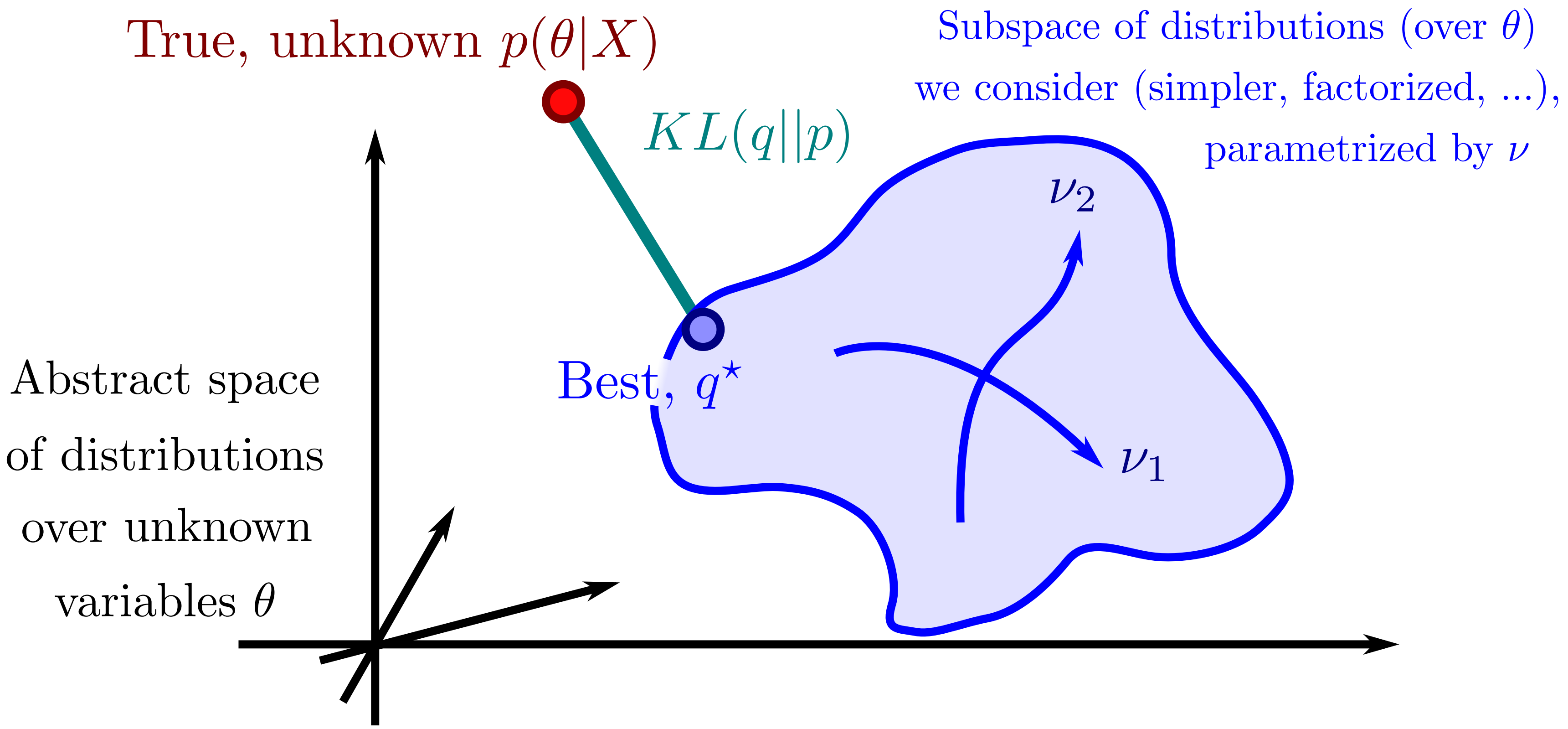
From EM to Variational Inference?

✐

Variational Inference

- Variational inference (VI)
  - basically, EM but more general
  - a kind of iterative and continuous version of importance sampling
- Variational Bayes (VB)
  - fully bayesian VI
  - same treatment for parameters θ and hidden variables z
  - i.e., $q(z, \theta)$ instead of $q(z)$ and θ_{ML}
- Useful for
  - any model, even without conjugate prior
  - changing the problem into an optimization problem (no sampling)
- Mechanisms
  - variational distribution q, variational parameters ν
  - minimizing the $KL\left(q(z, \theta) || p(z, \theta | X)\right)$
  - (maximizing the *ELBO* (evidence lower bound) (via Jensen's inequality))

Minimizing the KL?



Variational Inference: approach

- Have a fully bayesian probabilistic model
- Suppose additional independance (or approximation)
  - e.g., with $\theta = \{\mu, \sigma\}$ we could suppose $q(\theta) = q_1(\mu)q_2(\sigma)$
- Write analytical formula of $q^\star$ (with parameters ν)
  - e.g.
    - $q_1^\star(\mu) = q_{(\nu)}{}_1^\star(\mu) = \mathcal{N}(\mu|\nu_1, \nu_2)$
    - $q_2^\star(\sigma) = q_{(\nu)}{}_2^\star(\sigma) = \mathcal{N}(\sigma|\nu_3, \nu_4)$
  - in case of (supposed) independance (*mean field approximation*), this can be found by indentifying a distribution using
    - $q_1^\star(\mu) \propto e^{\mathbb{E}_{q_2^\star}[\ln p(\mu,\sigma,X)]}$
    - (or, in log-space)
    - $\ln q_1^\star(\mu) = K + \mathbb{E}_{q_2^\star}[\ln p(\mu, \sigma, X)]$
- Iteratively optimize (or GD) the parameters

ELBO: Maximize the Evidence Lower bound?

$$
\begin{aligned}
KL(q \,||\, p) = KL(q(\theta) \,||\, p(\theta|X)) &= \mathbb{E}_q\left[\ln \frac{q(\theta)}{p(\theta|X)}\right] = \int_\theta q(\theta) \ln \frac{q(\theta)}{p(\theta|X)} \\
&= \mathbb{E}_q\left[\ln \frac{q(\theta)p(X)}{p(\theta, X)}\right] \\
&= \mathbb{E}_q\left[\ln \frac{q(\theta)}{p(\theta, X)}\right] + \mathbb{E}_q\left[\ln p(X)\right] \\
&= \underbrace{\mathbb{E}_q\left[\ln \frac{q(\theta)}{p(\theta, X)}\right]} + \ln p(X) \\
&= \underbrace{\mathbb{E}_q\left[\ln q(\theta)\right] - \mathbb{E}_q\left[\ln p(\theta, X)\right]}_{-ELBO} + \underbrace{\ln p(X)}_{\text{normalization constant}}
\end{aligned}
$$

$$evidence = ELBO + KL$$
$$evidence \leq ELBO$$

- we can rewrite the "evidence" $\ln p(X)$ and use the Jensen's inequality[W] ($\ln$ being concave, $\ln \mathbb{E} \geq \mathbb{E} \ln$)

$$
\underbrace{\ln p(X)}_{\text{evidence}} = \ln \int_\theta p(X, \theta) = \ln \int_\theta p(X, \theta) \frac{q(\theta)}{q(\theta)} = \ln \mathbb{E}_q\left[\frac{p(X, \theta)}{q(\theta)}\right] \quad \underbrace{\geq}_{\text{lower bounded by}} \quad \underbrace{\mathbb{E}_q\left[\ln \frac{p(X, \theta)}{q(\theta)}\right]}_{\text{ELBO}}
$$

Proof that $q_1^\star(\mu) \propto e^{\mathbb{E}_{q_2^\star}[\ln p(\mu,\sigma,X)]}$

- Reminding our example with $q(\theta) = q(\mu, \sigma) = q_1(\mu) q_2(\sigma)$
- Using the calculus of variations[W] (hence the name "variational inference")
  see also https://wiki.inf.ed.ac.uk/twiki/pub/MLforNLP/WebHome/bkj-VBwalkthrough.pdf
- Sketch, approximate proof: we set the (partial) derivative of the ELBO to 0

Variational Inference for a 1d Gaussian[W]

- The "basic[W]" (and useless) case

✏

Variational Inference for GMM[W]

- The usual[W] case

Going beyond plain variational inference

- In the next part
  - Stochastic VI
  - Doubly Stochastic VI
  - Amortized Variational Inference
  - VAE (and beyond)
    - Wasserstein auto-encoders
    - β-VAE
    - Disentangling-VAE
    - ...
- and more...
  - Black-box VI
  - Generalized Variational Inference
  - good "tutorial"

Overview

- (Gaussian) Mixture Models (GMM)
- Expectation Maximization for GMM
- Variational Inference,

  Variational Bayes